Health & Technology
By Convergence

Title: Convergence Imaging Facility and Innovation Centre (CIFIC)

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

*Name**	*Institution*	*Department*	*Email*	*Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)*	*For Erasmus MC applicants: clinician/ researcher/ mix*	*Role in Flagship [Main Lead/Lead/ Applicant]*
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Planned start date of seed funding: *1-10-2022*

Primary Convergence theme: *Fundamentals of Health & Disease*

Secondary Convergence theme (*if applicable*): *Human Centered Technology and AI for Health*

Tertiary Convergence theme (*if applicable*): Choose an item.

Quaternary Convergence theme (*if applicable*): Choose an item.

Keywords: *light microscopy, electron microscopy, ultrasound microscopy, multi-modal multi-scale microscopy, image analysis, artificial intelligence, molecular mechanisms of disease*

1.b Summary (max 300 words: 296)

We aim to create a convergent imaging facility and innovation center (CIFIC) that accelerates the pace of innovation and application in bio-imaging and microscopy. CIFIC brings an integral approach, where innovation in microscopy is delivered by a flagship facility that provides both beyond-state-of-the-art equipment and an expertise and innovation ecosystem. This unique combination of technical and theoretical expertise (TU Delft), biomedical expertise (Erasmus MC) and innovation management and initiative (EUR) has several key advantages and opportunities:

1. It creates an extensive mutual feedback-loop between technology development and application at the various Technology Readiness Levels.
2. It enables optimization of an entire workflow, from sample preparation protocols such as tissue preservation and clearing, molecular probes and contrast agents, a set of interconnected instrumentation hardware, to computational algorithms and data analysis techniques tailored to specific biomedical applications.
3. It stimulates commercialization and market readiness by continuous feedback along the entire innovation and Technology Readiness Level (TRL) chain. The proposed way of working is also an attractive value proposition for industrial partners to set up collaborations and cultivate entrepreneurial activities originating from the work.

Interactions between the TU Delft and Erasmus MC groups (mutual feedback loops) will be catalyzed by the Optical Imaging Centre (OIC), a core facility (CoFa) at Erasmus MC, with an excellent reputation for translating fundamental, translational and (pre)clinical research questions into specific microscopy applications. This catalyzing role will be strongly enhanced by close collaboration with the Van Leeuwenhoek Laboratory for Advanced Imaging Research (VLLAIR) at TU Delft, which hosts a range of microscopy pioneers targeting fundamental technology breakthroughs and instrumentation innovation co-developed with industrial partners. Innovation and commercialization will be further accelerated via support from the Erasmus Center for Innovation, whose members are specialized in collaboration creativity and learning management in large multi-party innovation networks.

1.c Public Summary (max 300 words: 274)

Microscopie is al eeuwenlang een sleuteltechnologie voor het ontrafelen van de geheimen van het leven en van ziekteprocessen. Tegenwoordig worden geavanceerde microscopische technieken zelfs nog intensiever dan voorheen gebruikt voor fundamenteel, translationeel en (pre)klinisch onderzoek.

Ons doel is om de effectiviteit en specificiteit van technologische innovatie te maximaliseren, niet alleen van microscopen, maar ook van nieuwe labelingsmethodes en apparatuur om cellen en weefsels in leven te houden terwijl ze worden bestudeerd. Dit willen we bewerkstelligen door de krachten te bundelen van technologen en uitvinders van de TU Delft en onderzoekers van het Erasmus MC. De interactie tussen die twee partijen, de wederzijdse feedback-loop, zal worden gekatalyseerd en gekanaliseerd door het Optical Imaging Centre (OIC, erasmusoic.nl), een Erasmus MC Core Facility (CoFa) waarvan de participerende leden niet alleen uitstekende technologische expertise hebben, maar ook een achtergrond in het biomedisch onderzoek (PhD-niveau).

In het project zullen we de verschillende modules binnen de werkketen, van preparatie en labeling tot het ontwerp en gebruik van instrumenten en methodes voor data-analyse, koppelen aan een reeks uiteenlopende biomedische onderzoeksvragen met betrekking tot diverse ziektebeelden. Deze simultane benadering maakt het mogelijk om de gehele werkketen geïntegreerd te optimaliseren. Dit zal uiteindelijk leiden tot nieuwe medische inzichten en verbeterde diagnose en behandeling. Bovendien wordt hierdoor ook de technologietransfer naar commerciële samenwerkingspartners kansrijker en gemakkelijker.

Onze aandacht gaat in het bijzonder uit naar technieken die grote afgebeelde volumes (gezonde of zieke weefsels) combineren met een hoog oplossend vermogen, en naar technieken om daarin zelfs op de nanometer-schaal te kunnen kijken, de schaal waarop de moleculaire basisblokken van de cel op uiterst complexe manier samenwerken, of juist niet, zoals in het geval van ziekte.

2. Relevance to Convergence Health & Technology mission (50%) – max. 2500 words for 2.a to 2.e: 2472

• The Flagship program contributes to the Convergence Health & Technology mission (1)
• The Flagship program objectives fit with the guidance per theme provided by the theme leads, and/or fit with more than one theme of Convergence Health & Technology (2)
• The Flagship program merges approaches and insights from distinct disciplines to reach its goals (3)
• The staff granted by this call actively brings together different disciplines within the program, and creates visibility and attracts talents, partners, and funding for the program (4)
• The Flagship program has a clear view on long term sustainability after the 5-year central funding period (5)
• The envisioned results of the Flagship program have the potential for a significant contribution to healthcare, society and/or have scientific impact which has the potential for long-term societal impact (6)
• The Flagship program provides opportunities for BSc/BA and/or MSc/MA students of the participating institutes to participate in (interdisciplinary) projects (7)
• The proposal defines the pathway of how this contribution can be realized. The pathway includes concrete research and/or impact pathway (societal and/or economical) valorization strategies (8)

2.a Contribution to H&T mission and fit with H&T theme(s)

The research topics of the Erasmus MC participants are focused on revealing molecular mechanisms of cellular (dys)function in healthy and diseased cells and the response of the latter to therapy. The fundamental and translational research heavily relies on microscopic and sample preparation technology and will therefore strongly benefit from innovation and improvement of microscopy related technology with TU Delft. Technology transfer, at the heart of our proposal, enables discovery and invention to go hand in hand. The contribution to Convergence H&T is excellent, as the accelerated research output ultimately will be translated into disease models and various applications in the clinic, including therapy optimization and next generation diagnostics with molecular pathology.

The flagship fits with the items ‘Prevention and early diagnostics to proactively maintain health’ and ‘individual-tailored precise medical treatment for all’. The combination of a number of research groups that study different disease models with a wide array of microscopic techniques and the experts from TU Delft on microscopic technology will vastly enhance the possibilities for early diagnosis as well as for (personalized) treatment.

2.b Synergy between disciplines

The researchers in the Erasmus MC team intensively use various types of microscopy in their research focused on different disease model systems. Over the years they have developed qualitative and quantitative assays and read-outs specifically suited to answer their own research questions. However, the researchers are from different departments, and hence the communication between them has until now been limited, in turn limiting the *generic* application of their approaches.

At TUD, development of novel generic microscopy methods and assays is usually carried out only until proof-of concept, leading to delays in application- and market-readiness as workflow solutions are only addressed after development. Necessary mutual feedback between invention and actual application is often limited or even absent.

In the program we will tackle these problems by, **1)**, the formation of couples of Erasmus MC and TUD participants. Coupled researchers will *share* a post-doc responsible for linking their existing externally funded application-, resp. instrumentation-oriented research projects. Importantly, the shared post-doc project will, by mutual feedback during the developmental stage, significantly enhance the *query-driven* design and implementation of novel dedicated microscopy assays, and will also boost insight into microscopy in Erasmus MC and insight in biomedical research in TUD.

Moreover, we will, **2)**, build a grassroots community by fostering interactions, e.g. by our bi-monthly seminar cycle and annual science day, and also within our broader ecosystem (see 4d and Figure 1), with presentations, discussions, and demonstrations. This will not only enhance exchange of ideas, but is also expected to inspire individual researchers and couples to implement approaches developed by others and adapt them to the specific demands of their research. This will lead to more generic workflows for targeting the different disease models and research questions. Moreover the TUD participants will be able to identify opportunities to improve and adapt, and also to combine various modes of microscopy used in the different research lines, while Erasmus MC participants may be triggered to multi-modal approaches that offer higher content data.

Finally, we will **3)** use our existing ties to external users and the expertise of the EUR-RSM participants to accelerate translation of demonstrated microscopy workflows into actual products for the broader, global community of health researchers. Technical and social elements are strongly interwoven in determining the successful valorization of pioneering discoveries. RSM participants, mediated by a post-doc, will investigate interactions and innovation approaches both within the entire CIFIC community and in the shared post-doc projects to uncover successful strategies for collaboration in a unique query-driven multi-party collaborative network. Although inter-organizational collaboration initiatives in an innovation ecosystem have long been part of the health and technology management fields, there are many challenges related to managing these initiatives, which remain poorly understood. Uncovering the role of collaboration, creativity, and learning in joint innovative activities is an urgent matter not only to build new theory on convergence in health and technology, but also to strengthen the sustainability of CIFIC and to inform practice and policy in general. The generated knowledge will in turn allow further optimization of mutual feedback between research and query driven technology development, spearheading fast-track innovation and valorization.

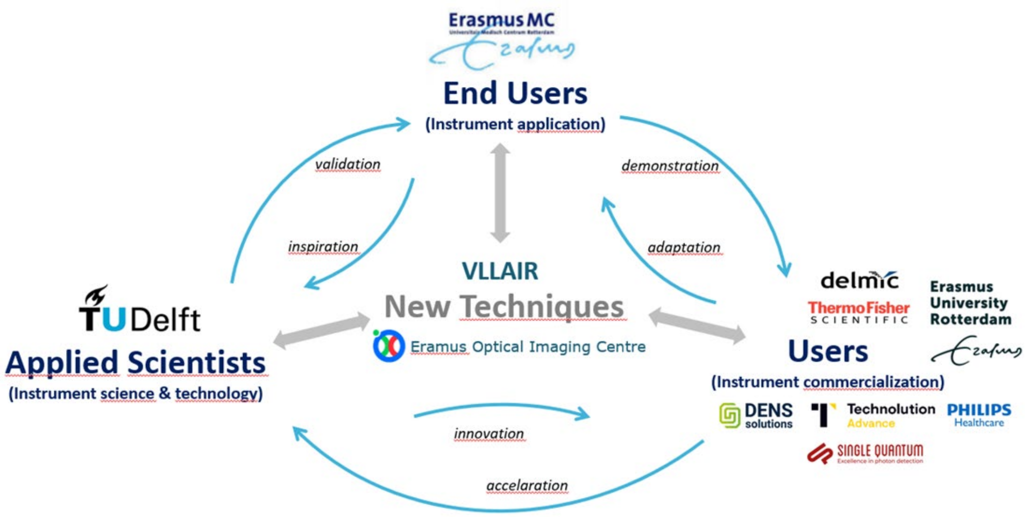

Figure 1: *CIFIC ecosystem in which microscopy facility connect clinical and pre-clinical end users of instrumentation to innovators and technology developers.*

2.c Long term sustainability plan

Feasibility

The feasibility of the program is particularly strong **first** of all by the creation described in Work Package 3 of in total 12 teams of Erasmus/TUD researchers and assigning a one- or two-year post-doc to each team. This has three main advantages: **1)** each participant of Erasmus MC-team has specific participant(s) of the TUD-team as a 'convergence anchor person', and vice versa; **2)** each post-doc has a well-defined research plan and corresponding task within the program, and the anchor persons will act as direct supervisors; **3)** each team can adapt its research and imaging applications to the microscopy development described in WPs 1 and 2, in which the TUD-team focuses on innovation and novel development of a large array of different, often complementary microscopic technology. A **second** pillar to our feasibility is the arrangement of TUD microscopy developers in WP 1 and 2 in two generic complementary multi-modal lines: 'Google Patient' and 'Ultra-resolution' (see 3c). Interactions within these WPs will be stimulated towards projects and development of multi-modal solutions and workflows, leading to technological challenges overarching the duration of the current funding round. **Third**, we fill a missing link in our combined expertise, by hiring a **technician/analyst** specialized in multi-modal and EM sample preparation at the OIC Core Facility and allocating this person 0.4 fte at TUD. A **fourth** and final contributor to feasibility is the strong connection to and involvement of a company network that is highly interested in participation in CIFIC and is willing to contribute with cash and in-kind contributions to future convergent project proposals, boosting our success to find future external matching via public-private partnership funds.

Long-term sustainability

All factors above also contribute to the sustainability of our program after the first five years funding period. The organization model, building on the well-defined collaborations of the specific couples and the frequent interactions between couples, provides an excellent base on which novel projects can be developed, and new grant proposals can be written. This will be initiated already during execution of the program, for instance by the post-docs under guidance of their PI-anchors. Moreover, the matching external projects of all PI's in the program will strongly benefit from our set-up, since the post-docs will work in narrow collaboration with the PhD-students and post-docs in these already funded projects. Also here, new ideas and approaches originating from the many cross-connections created in the program, will stimulate and help all PI's to write new innovative grant proposals, not only individually, but importantly, also in small groups or even in newly formed, possibly public-private, consortia. As the Dutch microscopy landscape is globally unique in terms of both academic and industrial breadth and our researchers are well immersed in this network, we foresee a strong competitive edge for CIFIC also in international funding schemes.

An additional factor is the substantial training of junior PIs and students in convergent thinking and providing them a convergent network. Our consortium consists of a large number of junior PIs, including 8 tenure-track, to which the Imaging Physics department intends to add an additional tenure-track PI from own resources to specifically work on microscopy approaches for clinical applications. Also, we are well immersed in educational tracks (see 2d) and we plan to further stimulate convergent thinking and initiatives of early career researchers by providing small scale seed funds to couples of PhDs and/or PDs.

Risks and mitigation.
We identified the following risks:

- Ecosystem size: we form a large consortium (33 PIs) with an extensive additional industrial and societal network and ambition to grow. Members have widely different backgrounds and perspective which may complicate interactions and creative thinking. Three PIs from the RSM Innovation Management group are involved in CIFIC and will investigate interactions within CIFIC along three lines (see 3c-WP4), a post-doc will strengthen this team halfway, providing feedback and advice to further optimize our interactions.
- External matching: our aims need 1.2M€ matching over years 3-5. We already approached external industrial partners about CIFIC and received positive feedback and interest to support CIFIC research proposals (see support letters). Initial discussions between OIC and VLLAIR about CIFIC have already led to discussions on a NWO-TTW Perspectief program (~4M€) initiative related to 3D microscopy for medical applications and various other ideas (see also 2d). With CIFIC, we add post-docs to bridge existing, externally funded projects providing a further stimulus to our PIs to raise the matching funds
- Interactions between CIFIC projects: We propose 2-years post-doc projects to bridge technology and application projects, which may limit overlap and interaction between the postdoc projects. To encourage transfer of knowledge, we implement a stepped scheme (see 3c) ensuring overlap between experienced and starting post-docs. Actions of RSM investigations will further help identifying and transferring lessons learned.

2.d Visibility, attracting talents and new partners

Visibility
The visibility of the new flagship, if granted, will benefit strongly from the already existing visibility of the two imaging centers, VLLAIR and OIC. In fact, we plan to rebrand the centers to the Delft and Rotterdam branches of CIFIC. To enhance this even more, we will communicate CIFIC in the various courses and meetings that we already organize on a regular basis. In addition, we will adapt our running courses (PhD, MSc and BSc, see Table 1) to the new situation and invite each other in these courses to contribute to them. We will also point out where the interactions are strongest.

Independent of this, we aim to join and contribute to nationally coordinated symposia such as the Dutch chromatin meeting, Dutch Biophysics and Society for Microscopy (NVVM) meeting, NL-bioimaging, NEMI.

When the program has sufficiently developed we intend to organize a national symposium, as well as a renowned international conference on the topic of multi-modal multi-scale microscopy for health. Hereby, we aim to advertise the collaboration between technical an biological groups in the Netherlands and abroad.

Student engagement
We will, as already pointed out, also encourage BSc and MSc students in the courses we take part in (see Table 1) and in seminars and courses organized for the various research schools of TUD, Erasmus MC, and RSM to do Bachelor or Master traineeships in the groups of the participants,

supervised by the post-dos and PhD students involved. The RSM researchers are strongly involved in the MSc Medical Business Innovation- a new program launched between RSM and Erasmus MC. We will also encourage these students to develop their MSc thesis and the Idea Management and Implementing Innovations capstone project within CIFIC, playing a key role in connecting the technical aspects of the program with the development of the innovation ecosystem. Specifically, their work is expected to contribute to identifying factors and processes that support the commercialization and market readiness of the innovations generated within the program's framework.

We will establish an Impact Team focused on education as part of our governance structure (see 4d) for stimulating course activities, student demonstrations and advertising joint MSc projects.

Early career researchers

With our convergence funding, we will establish 12 PD positions, which we will brand as CIFIC Research Fellows, to bridge microscopy development and medical application projects. This will provide a unique experience encompassing methodological and technical knowledge on different complementary forms of microscopy as well as insight in research applications and medical relevance. We expect this to be a distinctive advantage either for starting an independent academic positon, or for becoming staff member in renowned imaging facilities elsewhere. To stimulate convergent thinking and initiative among our junior consortium members we will further establish seed grants for couples of PhDs and/or PDs to pursue their own convergent project ideas, which will also train them in proposal writing and fund raising.

Table 1: *Microscopy-related courses currently given by CIFIC PIs allowing student involvement*

Course code	Course title	Program	CIFIC instructors
NB1150/51/63/64	Labcourse Nanobiology A1/B1 A2/B2	BSc-NB	Geertsema
NB1120	Biomolecular programming	BSc-NB	Houtsmuller, van Cappellen
NB2402	Microscopy/Nanoscopy practice	BSc-NB	Houtsmuller, van Cappellen
NB2041	Optics and Microscopy	BSc-NB	Carroll
NB2121	Image Analysis	BSc-NB	Brinks, Carroll
NB3021	Optics and Applications in Nanobiology	BSc-NB	Carroll
NB4020	High-resolution Imaging	MSc-NB, BME	Hoogenboom, Jakobi, Rieger
TN1101	Technology Management	BSc-AP, RSM	ImPhys & RSM staff
AP3531	Acoustical Imaging	MSc-AP	Maresca
AP3401	Introduction to charged particle optics	MSc-AP, PhD, external	Mohammadi-Gheidari, Hoogenboom
AP3132	Advanced Digital Image Processing	MSc-AP	Rieger
ONWAR / MolMed	OIC Imaging Course	PhD	Houtsmuller, van Cappellen
	Imaging Course	MSc-I&I	Houtsmuller, van Cappellen

NB: Nanobiology, AP: Applied Physics, BME: Biomedical Engineering, I&I: Immunity & Infection, ONWAR: Onderzoeksschool Neurowetenschappen Amsterdam-Rotterdam; MolMed: Molecular Medicine

Attracting talent, partners, and funding

Our strong involvement in the educational programs (see Table 1) allows screening for talented MSc students to pursue PhD positions within CIFIC. We will advertise PhD and PD positions using the CIFIC brand and consider introducing a rolling call for PhD and PD positions with CIFIC PIs. Our 33 PIs are frequently speaking at international conferences, which we will use to raise awareness for positions within CIFIC in general.

We have already gathered a community of industrial partners that see great added value in our strategy of query-driven microscopy innovation and who have indicated interest in contributing to public-private partnership projects and programs (*see support letters*), which will boost our chance to attract external funding. Involvement of RSM will allow further expansion of this network. We will also establish an Impact Team on ecosystem and societal partners (see 4d), responsible for maintaining and enlarging our network, specifically also towards societal partners such as patient organizations with which some of our Erasmus MC participants already have contacts. Concrete actions for future funding that we will pursue: a large NWO-TTW Perspectief consortium proposal on 3D microscopy development for health applications together with several of our industrial and external partners (expected size ~4M€), collaborative proposals within WP1,2 (e.g. NWO-TTW Open Technology Program) on digital pathology (~500k€), on cryo-correlative superresolution FLM-EM (~800k€), on large-scale data acquisition and analysis for light-sheet microscopy and multi-beam EM (~800k€), and collaborative proposals aimed at applications within WP3 (e.g. via ZonMW and NWA).

2.e Long term societal impact

The impulse to biomedical research generated by mutual feedback between TUD and Erasmus MC groups involved in query-driven innovation, development and application of microscopy related technology is expected to have a considerable scientific impact. Since the expansion of biomedical knowledge is expected to be accelerated by the flagship interactions the final societal impact will be seen in improved diagnosis and (personalized) therapy. Economic impact will be substantial through close partnerships with the particularly strong national and regional industrial ecosystem in life sciences instrumentation. Novel assays and equipment will boost spin-offs like Delmic (Hoogenboom, TUD flagship lead), or companies within the applicants' networks like Thermo Fisher, Leica, Zeiss, Technolution, Single Quantum, Demcon, and DENSsolutions.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d: 1961

• The Flagship program contributes to excellent Convergence Health & Technology research, at the forefront of international research (1)
• The Flagship program is both innovative as well as challenging in its research and approach (2)
• The Flagship program has a clear strategy to engage students in its research plan (3)
• The Flagship program has a clear aim and long-term work plan, including, if applicable, a specification of required facilities and a data-and infrastructure plan. Coherence within the program is clearly demonstrated (4)
• The Flagship program aligns with or builds on (inter)national research agendas in this field, fit with the (inter)national research agendas but has a "high risk, high gain" main objective that is thoroughly explained (5)
• The Flagship program is feasible in terms of research performed, staff and budget needed (6)
• The Flagship program has a concrete plan for attracting external funding, which should result in at least the same amount as the central funding over the 5-year period (7)

3.a Ambitions

Innovations in microscopy have been driving scientific progress over the past centuries. Microscopy data is key to obtain insight in the fundamental processes of life and disease. Present-day challenges require data with higher information content and in larger quantity than what can be obtained now. Current imaging facilities have limited access to beyond-state-of-the-art technology, and inventors of novel technology need better feedback of biomedical researchers.

We are convinced that creating mutual feedback loops between development/invention/ innovation at TU Delft and biomedical research in Erasmus MC will generate new scientific discoveries and enhanced effectiveness of therapies.

3.b Aims and feasibility

Our aim is to create a multi-modal, multi-scale microscopy platform focusing on novel instrumentation and correlative workflows for biomedical use cases. These workflows are assembled from several generic building blocks and can be tailored to specific applications. The building blocks fall in categories of (i) sample preparation, probes and labels, (ii) imaging hardware including various forms of light and electron and sound microscopy, (iii) computational methods for image analysis and processing. Together they will form a library of workflow building blocks, ready for matching to applications at hand.

Students are involved *de facto*, as the proposed research in 3c is fully based on the current and future matching PhD- and MSc projects in the groups of the participants. We next propose to appoint convergent CIFIC post-doc fellows who will be bridging the technical development project to the application project while being fully integrated with both these projects. This approach also warrants the feasibility in terms of research performed, staff, and budget need (see also 2b, 2c, and 3c), paving the way for further convergent proposals by all project partners involved.

We further aim to accelerate innovation and market entry of the new technology through direct involvement of an extensive network of industrial partners as well as the Innovation Management group at Rotterdam School of Management to investigate optimization and management of innovation activities (see 2b, 4c, support letters).

3.c Work plan

We have defined four connected work packages. WP1 and WP2 concern the development of microscopy techniques (including probes, sample preparation, data handling, and computation) as individual building blocks (BBs) and the connection of these BBs into generic correlative workflows (see Fig.1, 2). With WP1 and WP2, BBS are aligned along two workflows; 'Google Patient' (WP1), imaging from the molecular scale to organ or organism level; and 'Ultraresolution' (WP2) targeting the molecular and atomic scale structural layout of macromolecular complexes in cells and tissue. WP3 connects key biomedical applications to novel technology based on the BBs developed in WP1,2 (see Fig.3). WP4 concerns collaboration, creativity, and learning in innovation projects to inform and optimize current work as well as future collaboration efforts among innovation partners in the CIFIC ecosystem.

WP 1 – 'Google Patient': from millimeters to nanometers

Dysregulation of molecular processes that control cellular function in healthy cells manifest in defects on the cellular, tissue, and organ(ism) level in disease. WP1 targets microscopy techniques and multi-modal workflows along all these scales. 3D fluorescence light microscopy (FLM) is key for visualizing dynamics of molecular and cellular processes, but needs higher resolution and higher throughput to bridge molecular, cellular and organism scales. Optogenetic and other novel probes provide functional read-outs, which need combination with optimized 3D FLM design to facilitate multi-scale functional imaging. Higher throughput is also needed to allow targeting multiple individuals and assess interpersonal variability to develop therapeutic strategies for personalized medicine. Correlation with electron microscopy (EM) is necessary to provide structural context that remains invisible with FLM. Multi-beam scanning EM stands to increase throughput 100x, matching FLM fields of view and generating multi-scale correlative atlases ('visual omics'). Correlative and live 3D data analysis needs to be optimized and suitability for diagnostic purposes assessed. Finally, current diagnostic practice and evaluation of patient-derived material needs development of fast, 2D and 3D label-free microscopy methods.

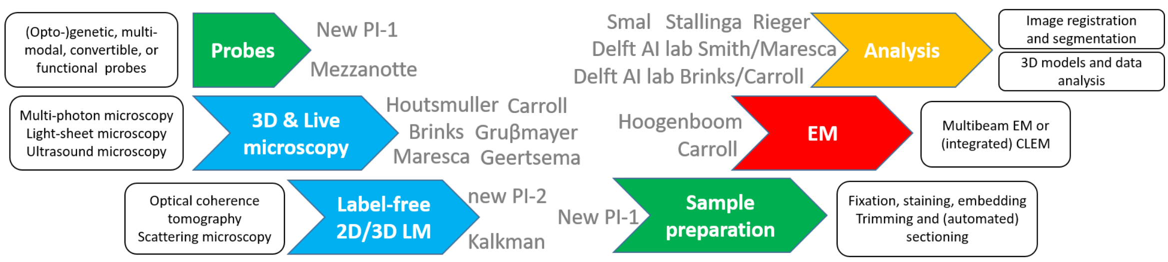

Figure 1: *Technologies within WP1 that constitute building blocks to target large-scale imaging with isotropic 1-100 nm resolution. Technological breakthrough is to generate data that allows zooming in and out across scales and content in a 'Google Maps' like fashion ('Google' Patient). This needs innovations within the individual building blocks as well as techniques to correlate data in research-question specific multi-modal workflows. Building blocks relate to sample preparation/probes (green), light or ultrasound microscopy (blue), electron or ion optics (red) and data handling and analysis (orange).*

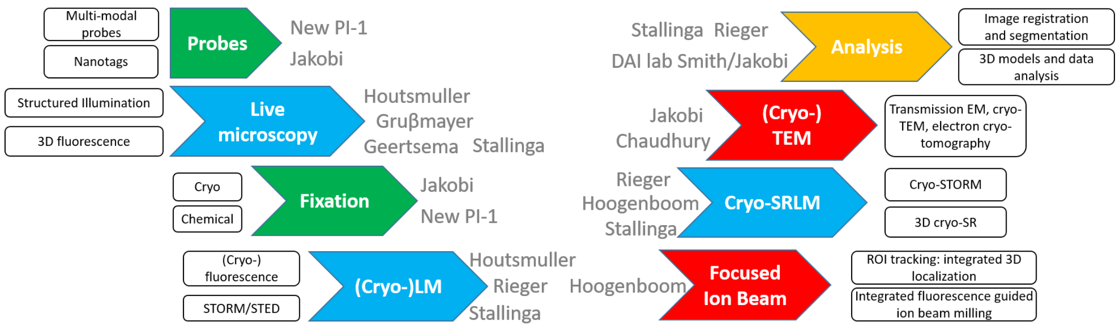

Figure 2: *Technologies within WP2 that constitute building blocks to target ultra-high resolution. Breakthrough technology will reveal proteomic content together with atomic-scale structural lay-out of macromolecular complexes in cells and tissues. Developed building blocks need incorporation in research-question specific multi-modal workflows. Colors similar to Fig.1*

WP2 – 'Ultra-resolution': multi-modal structural biology

EM and FLM are complementary and indispensable tools to visualize biomolecular machines within in cells, contributing essential knowledge for the design of new therapeutic strategies. EM at very low temperatures (cryoEM) allows zooming into the three-dimensional structure of macromolecules and their assemblies at almost atomic resolution in their native cellular context. Targeting biomolecules of interest requires 3D localization within tissue or cell. The ability to do so, critically depends on advanced FLM. Bridging these modalities and linking structural data to function therefore requires combined innovative workflows in (cryo)EM, LM at cryogenic conditions with super-resolution (cryoSRLM), larger scale live LM and non-cryo SRLM, cryogenic fixation and preservation of the sample in between imaging modalities, new and better probes for cryoSRLM and cryoEM, advanced algorithms for image reconstruction, smart data collection, and automation as well as cross-modality image registration and correlation.

WP3 – Biomedical applications: from state-of-the-art to beyond-state-of-the-art

In WP3 we couple **1)** specific research groups within the Erasmus MC that investigate various disease models and currently apply advanced state-of-the-art microscopy technology in the optical imaging centre (OIC), to **2)** TUD groups that develop beyond-state-of-the-art microscopic building blocks which will bring this biomedical research to a currently inaccessible level. **We will assign 11 post-docs to bridge existing (and future) biomedical research and technology development projects.** The post-docs will thus be embedded in the PhD/post-doc/MSc projects of TUD and ErasmusMC-PI's, creating larger convergent teams and providing a driver for external matching. In addition, the post-docs will be pivotal in creating the early mutual feedback between technology and application (see 1b, 2b). Also, **we will hire a technician/analyst within OIC specialized in probes and (multi-modal) sample preparation who will work 0.4fte at TUD to encourage generic sample preparation routines and exchange of samples**.

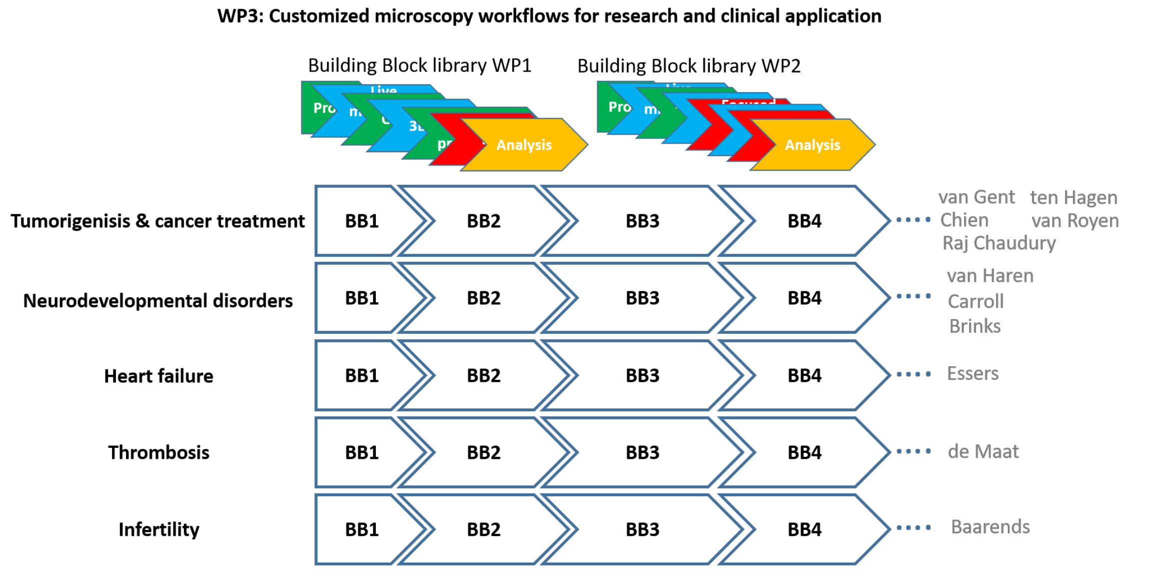

Figure 3: *Novel microscopy techniques, building blocks developed in WP1, 2, will be applied to key research questions and (pre-)clinical applications. Building blocks can be connected into multi-modal, multi-scale workflows required for specific research questions.*

<u>Biomedical research focusses on five different disease models:</u>

1. *Cancer* (van Gent, Chaudhuri, Chien, Ten Hagen, van Royen)
2. *Heart failure* (Essers)
3. *Neurodevelopmental disorders* (van Haren)
4. *Thrombosis* (de Maat)
5. *Infertility* (Baarends)

Below we briefly describe ErasmusMC research lines and indicate to which TUD-participants these will initially be coupled. Research lines are grouped within themes based on type of microscopy development in TUD. The post-docs assigned to each project will fully immerse in the described research and collaborate intensively with the PhD/MSc students involved.

<u>i. Multi-well light sheet microscopy (**Stallinga**) and multi-well microfluidics (**Ghatkesar**)</u>

- **van Gent** investigates the DNA damage response to chemotherapeutics of live patient tumor tissue by quantitative imaging in *microfluidic* devices. Ultimately, the goal is to determine which therapeutic is best suited for which patient (personalized medicine).

- **Essers** uses novel *in vitro* heart disease models to test the efficacy of drugs to treat symptoms of heart failure based on *microfluidic* technology. Prevalent genetic mutations are introduced with CRISPR/CAS. Quantitative imaging technologies in cardiovascular model systems will be developed for fast and sensitive detection of cardiac diseases.

- **Mezzanotte** develops novel imaging probes to enable understanding of tumor immunology leading to better immunotherapy. The team aims to develop multiplex bioluminescence microscopy methods to image live cells in *microfluidic* systems and the development of novel and brighter luciferase/luciferin variants.

ii. Correlative EM/light microscopy (**Hoogenboom/Jakobi**) and mechanical measurements (**Koenderink**)

- **Chaudhuri** aims to gain mechanistic insights into DNA replication stress and its role in tumorigenesis. *Electron microscopy* serves to analyze the architecture of replication intermediates. *High content quantitative fluorescent microscopy* adds analysis of protein recruitment dynamics at replication forks.
- **de Maat** aims to study architecture of the fibrin network in blood clots generated from thrombosis patient blood (using microfluidic devices) by *light and electron microscopy*. Methods to quantify features like density, fiber thickness and pore size are combined with *rheometric* determination of mechanical properties and sensitivity to lysis of the thrombi.

iii. Light-sheet and multi-photon microscopy with optogenetics probes (**Carroll/Brinks**)

- **van Haren** investigates how neurons navigate through the brain to connect to specific targets and how malfunction leads to neurodevelopmental disorders. The team will be developing micro-patterning devices and a microscope to use as a precision engineering tool to control biochemical processes at subcellular level, through tunable patterned illumination combined with optogenetics.

iv. Multiplane super resolution microscopy (**Grussmayer**)

- **Baarends** aims to unravel mechanistic aspects of homologous chromosome pairing during mammalian meiosis. They will use *superresolution* microscopy of meiotic cells retrieved from genetically modified mouse models, or through in vitro differentiation of embryonic stem cells using an organoid approach.

v. High-throughput microscopy for phenotype-based cell isolation and characterization (**Brinks**)

- **Chien** applies multidisciplinary technologies (microscopy, AI, and single cell technology) to investigate mechanisms of tumorigenesis. Microscopy-based cell profiling techniques are being developed, aiming to link tumorigenic phenotypes to genotypes. A dedicated microscope was developed with which features like individual cell phenotype can be acquired and analyzed to predict the fate of cancer-driving cells.

vi. Optical diffraction tomography (**Kalkman**)

- **van Royen** develops 3-D imaging and image analysis methods to improve diagnosis of prostate cancer. They developed protocols to optically clear clinical tissue to compare benign prostate lesions and malignant prostate tumors.

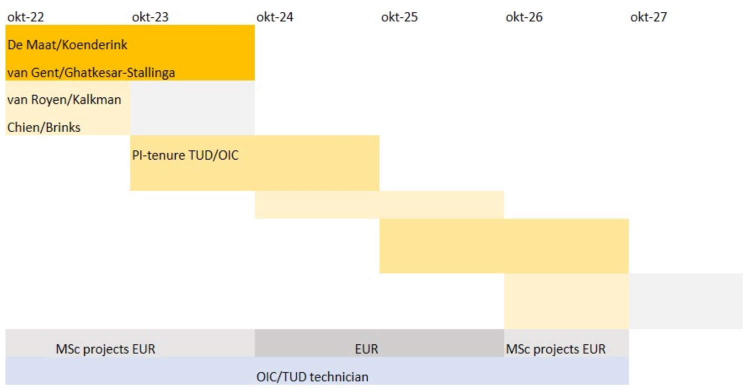

Figure 4: *Time schedule indicating employment of convergence personnel (post-docs and technician) over the granting period. Kick-starting projects in year1 have been indicated, others will be assigned as indicated above depending on progress in technology development*

WP4 – Accelerated innovation and collaboration

WP4 thus investigates optimization and management of innovation activities along three lines:

1. Interorganizational partnerships (**Tarakci**)
   How do partners collaborate within the convergence WP3-projects and in the CIFIC ecosystem as a whole? Aim is to uncover the mechanisms that enable multi-party innovation teams to become successful.

2. Creative problem solving (**Ananth**)
   Which processes and practices are implemented to foster creativity and the acceptance of new developments? Focus is on how creative ideas are uncovered and how scientists at TUD and Erasmus MC solve problems.

1. Experiential learning (**Madiedo**)
   What do project partners learn to further improve their innovation performance? Goal is to investigate how individual experiential learning processes support the innovation performance of the members of a growing ecosystem.

Research in the three lines is strengthened by a post-doc in year3 of our consortium and further carried out in MSc and Idea Management and Implementing Innovations capstone projects within the new ErasmusMC-RSM MSc Medical Business Innovation (see 2d).

3.d Alignment of the program with relevant national and international research agendas

OIC and VLLAIR are well embedded within **NWO large-scale research infrastructures roadmap** consortia NL-Bioimaging and NEMI and an extensive network of public-private partnerships, which is further boosted by participation of RSM (see 4c). We are well linked to the **High Tech Systems and Materials (HTSM) topsector**, with several matching projects supported by HTSM domain funding (resp. Kalkman, Hoogenboom, Stallinga). We further participate in the life sciences metrology program of the NXTGEN HighTech application for the **Nationaal Groeifonds**.

Multi-scale and multi-modal microscopy related research questions are explicitly mentioned in the **Knowledge and Innovation Agenda (KIA)** for Key Enabling Technologies (specifically ST1-2, ST3-5, ST4-1). Furthermore, our research topics are directly relevant for 8 routes on the **Dutch Research Agenda (NWA)** (1. Gezondheidszorgonderzoek, preventie en behandeling; 2. Meten en detecteren: altijd, alles en overal; 3. NeuroLabNL: dè werkplaats voor hersen-, cognitie- en gedragsonderzoek; 4. Personalised medicine: uitgaan van het individu; 5. De quantum / nano-revolutie; 6. Regeneratieve Geneeskunde: gamechanger op weg naar brede toepassing; 7. Smart industry; 8. Waardecreatie door verantwoorde toegang tot en gebruik van big data)

4. Consortium (25%) max 2000 words for 4.a to 4.d :1250

• The Flagship program is driven and executed by a well-organized team of scientists from all three institutions and at least 4 departments or faculties in total. In exceptional cases, it is possible to form a Flagship program with only two leads from two institutions, but this should be clearly motivated (1)

• Members of the consortium have a track record of excellence in academic research, translation of scientific results into clinical application or societal impact (2)

• Members of the consortium have a significant track record in engaging and supervising BSc and MSc students in research projects (3)

• Members of the consortium have demonstrated to be able to attract external funding, thereby enhancing the programs sustainability after the 5 years of central funding (4)

• There is a demonstrable fit between the aims and objectives of the program and the expertise of the consortium members (5)

• End users (private and/or public sectors) are involved from an early start and team members are well-connected to external partners in the ecosystem (6)

• Governance should be clearly described and is organized in such a way that it facilitates collaboration and optimally makes use of the expertise of all participants (7)

4.a Track record of the consortium members

It is the **core business** of all PIs in the program to obtain external funding for PhDs and postdocs, which they supervise and in which they are all successful. Many of these funded students are involved in projects that will kick-start CIFIC (external in cash matching, see budget table and support letters). A **non-exhaustive** list includes, HTSM 'Cryo3beams' (Hoogenboom/Jakobi; development of fluorescence-guided cryo-EM for WP2), EU-React 'ImDAP' (Hoogenboom/Carroll; development of image and data platform for WP1), NWO-Vici (Rieger; development of 1 nm cryo super-resolution LM, WP2), ERC-StG (Brinks; development of multi-photon voltage imaging microscopy for WP1) / Jakobi; development of cryo-TEM building blocks for WP2), NWO-NEMI FAST-EM (Hoogenboom; installation of multi-beam SEM at TUD, WP1), TKI (Mezzanotte; HH-TKI Oa-biodetectchip), CRC (ten Hage/Houtsmuller drug delivery by liposomes;WP3), NWO Building Blocks of Life (Houtsmuller work package; plasticity and dynamics of the genome structural organization), KWF (breast cancer-on-chip; van Gent/Houtsmuller; WP3), OnCode (Molecular mechanisms of meiosis; Baarends/Houtsmuller; WP3). Based on this past performance it is certainly expected that similar funding proposals will be granted in the future, maintaining and even expanding external matching as spin of the flagship program. Note that we will actively encourage PI's and post-docs involved for the flagship to write funding proposals during the execution of the program projects they are on.

As pointed out in detail in 3c, the research of the Erasmus MC participants focusses on various disease models. All their projects have specific components concerned with translation to clinical applications, as evidenced in their academic track records and publications (see selected references in Annex IV). The research of the VLLAIR participants has a strong focus and track record in valorization with spin-off companies like Delmic grown to maturation and several innovations successfully brought to the market, like the recent multi-beam FAST-EM.

It is also the core business of PI's in the program to teach in the various BSc and MSc curricula of the TUD, Erasmus MC and EUR (see table 1) and to continually supervise BSc and MSc students from these curricula, and also from other institutes in the Netherlands. In many labs of our PI's, BSc and MSc students from different disciplines (nanobiology, biomedicine, biomedical and mechanical engineering, and applied physics) collaborate within their projects. These students will also be involved in the planned frequent meetings with all PI's and post-docs.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

The project team is a balanced group of both TUD, EMC, and RSM senior scientists with strong track records and numerous publications in the top quartile of their fields. The team also includes young promising PIs (incl. 9 tenure-track PIs) who explore novel and innovative ways to develop novel technology (TUD), to use this to investigate health and disease at the fundamental, translational and preclinical level (Erasmus MC), and foster creativity and innovation in project teams (RSM).

The mutual feedback between technology innovation at TU Delft and application at Erasmus MC will be catalyzed by the members of the Erasmus Optical Imaging Centre (OIC). The OIC Core Facility has an excellent reputation for being able to bridge research questions and technology development. The seven members, of which four have a PhD in biomedical research and microscopy application and three are skilled technicians, train and support a variety of researchers within the Erasmus MC and often collaborate intensively with them, which has led and is continuing to lead to a large number of shared publications. Inter-organizational collaboration and application-driven innovation will be further boosted through involvement of world-renowned scholars on innovation management and innovation management related processes and practices at RSM. The expertise of the RSM PIs spans innovation strategy (Tarakci), creativity (Ananth) and learning (Madiedo). In addition to their academic publication records, the PIs disseminate knowledge and create impact by regularly helping a diverse set of organizations with their innovation and creativity-related challenges.

At TU Delft the group of participating PIs is well versed in a variety of sub-fields in microscopy, including functional microscopy (Brinks, Carroll), correlative light-electron microscopy (Hoogenboom), super-resolution microscopy and computational imaging (Smith, Rieger, Stallinga), OCT and large scale 3D microscopy (Kalkman), cryo-electron microscopy (Jakobi), and ultrasound microscopy (Maresca). There is a good mix in levels of seniority and in approach such that many different workflow building blocks can be advanced.

4.c Involvement of partners

Companies or start-ups are essential to bring innovations and new techniques to a larger user group outside our consortium, impacting health, science, and society globally. Providing the local Netherlands microscopy industry with revolutionary new products also impacts society via increased economic activity and employment. The Netherlands traditionally holds a strong position in microscopy and our consortium is very well connected to both academia and industry. Hoogenboom is co-founder and supervisory board member of **Delmic**, a 2010 TUD spin-off in the field of light and electron microscopy. The Delft EM groups also have traditionally strong ties with **Thermo Fisher Scientific**, leading manufacturer of electron microscopes, and Mohammadi-Gheidari holds a shared position between Thermo Fisher and TUD. Further companies with which our PIs have ongoing collaborations include **DENSSolutions** (Chaudhuri – holders for in-situ and high-resolution EM), **Single Quantum** (Zadeh – superconducting single photon detectors), **Technolution** (Hoogenboom, Smith – electronics and data solutions), and **Philips** (Tarakci, Stallinga – digital pathology, interventional guided therapy). The Innovation Management group of Tarakci further has a large company network via the **School of Management**. The OIC Core Facility (Houtsmuller) and several TUD-ImPhys researchers are well connected to major suppliers of light microscopy equipment, like **Zeiss**, **Leica**, and **Nikon**. Besides opportunities for economic valorization, this

network also provides **access to new beta-type microscopes** such as 64-beam FAST-EM and integrated cryogenic FM-FIB-SEM in the TUD VLLAIR labs.
Our PIs are also well immersed in the (inter)national academic networks. Hoogenboom is executive board member in **NEMI – Netherlands Electron Microscopy Infrastructure**, and has strong long-lasting collaborations with biomedical EM facilities at UMC's in Groningen (Giepmans), Utrecht (Klumperman), Leiden (Koster), and Maastricht (Peters). Houtsmuller and Rieger are Flagship leaders in **NL-Bioimaging AM**, the Dutch light microscopy collaboration, for superresolution and functional imaging microscopy. We are further participating in **Euro-Bioimaging** (via NL-Bioimaging) and **Instruct-ERIC** (Jakobi) and the researchers at ErasmusMC all are involved in various international consortia.

4.d Governance, organization and collaboration strategy of the consortium

Our virtual institute will be organized along two major axes: science and impact. The science axis will be structured via the teams that run the 4 WPs described above. The impact axis will be structured via convergent teams consisting of researchers from all three institutes. We will start impact teams devoted to:
(1) Developing and expanding our network with societal partners, such as different companies that provide, e.g., instrumentation, workflow solutions and AI methodology, as well as patient organizations and other stakeholders.
(2) Visibility & outreach, where we will start and exploit flagship social media accounts (twitter, tiktok, instagram), and where we will create inspiring microscopy and bio-imaging demonstrations for primary and/or secondary schools.
(3) Education, for setting up and advertising joint MSc-projects, and for maintaining a palette of MSc elective courses on (microscopy) imaging for pre-clinical and clinical research within the joint Nanobiology (TUD-EMC) and Medical Business Innovation (RSM-EMC) programs.

Overall coordination of the flagship will entail:
(1) setting up a bi-monthly faculty meeting for all our PIs, which will be the primary forum for democratic decision making.
(2) creating a yearly science day, focusing on the junior community of PhDs and PDs, and for bringing in international speakers.
(3) a bi-monthly seminar series, alternating with the faculty meeting.
(4) distributing small seed grants for couples of PhDs and/or PDs of different institutes, to start exploring new avenues, stimulate convergent thinking, and to practice independence from their respective PIs.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

IV References

Budget formulier: Convergentie Flagship Call

Flagship:	Convergence Imaging Facility and Innovation Centre
Flagship acronym:	CIFIC
Naam Penvoerder:	Jacob Hoogenboom
Startdatum en looptijd van het project:	1-10-2022 5 jaar
Totaal project budget:	> € 6.130.000

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.

Loonkosten:								2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maandbedrag	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																						
Organisatie 1,2 TUD/EMC aanvraag	postdoc	1,0	1-1-2023	31-12-2024	Schaal 10,5	€ 4.209	€ 151.519			€ 75.760		€ 75.760								€ 151.519	€ 0	€ 151.519,00
Organisatie 1,2 TUD/EMC aanvraag	postdoc	1,0	1-1-2023	31-12-2024	Schaal 10,5	€ 4.209	€ 151.519			€ 75.760		€ 75.760								€ 151.519	€ 0	€ 151.519,00
Organisatie 1,2 TUD/EMC aanvraag	postdoc	1,0	1-1-2025	31-12-2026	Schaal 10,5	€ 4.209	€ 151.519							€ 75.760		€ 75.760				€ 151.519	€ 0	€ 151.519,00
Organisatie 1,2 TUD/EMC aanvraag	postdoc	1,0	1-1-2025	31-12-2026	Schaal 10,5	€ 4.209	€ 151.519							€ 75.760		€ 75.760				€ 151.519	€ 0	€ 151.519,00
Organisatie 1 TUD matching	Tenure Tracker	1,0	1-10-2022	1-10-2027	Schaal 11	€ -			€ 23.004		€ 92.015		€ 92.015		€ 92.015		€ 92.015		€ 69.011	€ 0	€ 460.074	
Organisatie 2 EMC aanvraag	Technicus/analyst	1,0	1-10-2022	1-10-2027	Schaal 8	€ 3.863	€ 353.465	€ 17.673		€ 70.693		€ 70.693		€ 70.693		€ 70.693		€ 53.020		€ 353.465	€ 0	€ 353.464,50
Organisatie 3 Erasmus Universiteit aanvraag	postdoc	1,0	1-1-2024	31-12-2025	Schaal 10	€ 4.209	€ 151.519					€ 75.759		€ 75.759						€ 151.519	€ 0	€ 151.518,96
Organisatie 1,2 TUD/EMC aanvraag	postdoc	1,0	1-1-2026	31-12-2027	Schaal 10	€ 4.209	€ 151.519									€ 75.759		€ 75.759		€ 151.519	€ 0	€ 151.518,96
Organisatie 1 TUD aanvraag	postdoc	1,0	1-1-2024	31-12-2025	Schaal 10	€ 4.209	€ 151.519					€ 75.759		€ 75.759						€ 151.519	€ 0	€ 151.518,96
Organisatie 1,2 TUD/EMC aanvraag	postdoc	1,0	1-1-2026	31-12-2027	Schaal 10	€ 4.209	€ 151.519									€ 75.759		€ 75.759		€ 151.519	€ 0	€ 151.518,96
Organisatie 1,2 TUD/EMC aanvraag	postdoc	1,0	1-1-2027	31-12-2027	Schaal 10	€ 4.209	€ 75.759											€ 75.759		€ 75.759	€ 0	€ 75.759,48
Organisatie 1,2 TUD/EMC aanvraag	postdoc	1,0	1-1-2027	31-12-2027	Schaal 10,5	€ 4.209	€ 75.759											€ 75.759		€ 75.759	€ 0	€ 75.759,48
Organisatie 1,2 TUD/EMC aanvraag	postdoc	1,0	1-1-2027	31-12-2027	Schaal 10	€ 4.209	€ 75.759			€ 75.759										€ 75.759	€ 0	€ 75.759,48
Organisatie 1,2 TUD/EMC aanvraag	postdoc	1,0	1-1-2027	31-12-2027	Schaal 10	€ 4.209	€ 75.759			€ 75.759										€ 75.759	€ 0	€ 75.759,48
TOTAL personeel aanvraag							**€ 1.868.654**	**€ 17.673,23**	**€ 23.003,70**	**€ 373.730,86**	**€ 92.014,80**	**€ 373.730,86**	**€ 92.014,80**	**€ 373.730,86**	**€ 92.014,80**	**€ 373.730,86**	**€ 92.014,80**	**€ 356.057,60**	**€ 69.011,10**	**€ 1.868.654,26**	**€ 460.074,00**	**€ 1.868.654,26**
Ondernemingen																						
Onderneming 1						€ -	€ 0													€ 0	€ 0	€ 0,00
Onderneming 2						€ -	€ 0													€ 0	€ 0	€ 0,00
Onderneming 3						€ -	€ 0													€ 0	€ 0	€ 0,00
Onderneming 4						€ -	€ 0													€ 0	€ 0	€ 0,00
Overige (private) partners NWO/EU/MD (MATCHING VANUIT HUIDIGE PROJECTEN)																						
THHC12 NWO CRYO3BEAMS	postdoc	1,00	1-10-2020	1-10-2022	Schaal 10,5	€ -		€ 17.464		€ 5.821										€ 23.285	€ 0	€ 23.284,50
THHC12 NWO CRYO3BEAMS	postdoc	0,80	1-5-2020	1-4-2023	Schaal 10,5	€ -	€ 0	€ 13.971		€ 13.971										€ 27.942	€ 0	€ 27.941,75
THHC15 ERC MULTIVISION	postdoc	1,00				€ -	€ 0	€ 5.641		€ 67.692		€ 62.051								€ 135.384	€ 0	€ 135.384,00
THHC15 ERC MULTIVISION	Phd	1,00	4-2-2020	4-2-2024		€ -	€ 0	€ 12.387		€ 49.548		€ 4.129								€ 66.064	€ 0	€ 66.064,00
THHC15 ERC MULTIVISION	Phd	1,00	1-9-2020	1-9-2024		€ -	€ 0	€ 12.387		€ 49.548		€ 33.032								€ 94.967	€ 0	€ 94.967,00
THHC18 NWO Biomolecular 'Google Maps'						€ -	€ 0													€ 0	€ 0	€ 0,00
THHC21 Imaging Data Platform	postdoc	1,00	15-9-2021	15-9-2023		€ -	€ 0	€ 35.517		€ 100.632										€ 136.149	€ 0	€ 136.148,50
THHC21 Imaging Data Platform	postdoc	1,00	15-12-2021	15-12-2023		€ -	€ 0	€ 142.068		€ 136.149										€ 278.217	€ 0	€ 278.216,50
THHC22 TKI MEMS-based electron microscopy	postdoc	1,00	1-10-2022	1-10-2024		€ -	€ 0	€ 35.517		€ 142.068		€ 106.551								€ 284.136	€ 0	€ 284.136,00
THHC23 ZonMW Pancreatic exocrine malfunction						€ -	€ 0													€ 0	€ 0	€ 0,00
THMD01 NWO Ultra-resolution with visible light	postdoc	1,00				€ -	€ 0			€ 37.122		€ 37.122								€ 74.244	€ 0	€ 74.244,00
THMD01 NWO Ultra-resolution with visible light	Phd	1,00	1-12-2019	30-11-2023		€ -	€ 0	€ 13.130		€ 48.142										€ 61.272	€ 0	€ 61.272,17
THMD01 NWO Ultra-resolution with visible light	Phd	1,00	1-9-2020	1-9-2024		€ -	€ 0	€ 13.327		€ 53.307		€ 35.538								€ 102.172	€ 0	€ 102.171,75
THMD01 NWO Ultra-resolution with visible light	Phd	1,00	1-5-2021	1-5-2025		€ -	€ 0	€ 13.130		€ 52.519		€ 17.506								€ 83.155	€ 0	€ 83.155,08
THMD04 NWO Computational imaging	postdoc	1,00	1-6-2022	1-12-2024		€ -	€ 0	€ 17.494,80		€ 69.979,20		€ 64.147,60								€ 151.622	€ 0	€ 151.621,60
THFC80 NWO Biomolecules for acidity using ultrasou	PhD	1,00	15-12-2019	14-12-2023		€ -	€ 0	€ 15.298,25		€ 61.193,00		€ 56.093,58								€ 132.585	€ 0	€ 132.584,83
THFC83 EU Multi Messenger Imaging of Cancer	postdoc	1,00	1-1-2022	1-1-2024		€ -	€ 0	€ 17.596,50		€ 70.386,00										€ 87.983	€ 0	€ 87.982,50
TKI Health Holland & Stichting Dierproefvrij	Volledige projectkosten		1-1-2021	31-12-2023		€ -	€ 0	€ 21.500,00		€ 86.000,00										€ 107.500	€ 0	€ 107.500,00
NWO Stop and Start, system wide analysis to transcri	Volledige projectkosten		1-10-2019	30-9-2024		€ -	€ 0	€ 75.000,00		€ 300.000,00		€ 225.000,00								€ 600.000	€ 0	€ 600.000,00
NWO Modulating DNA replication stress response	Volledige projectkosten		1-10-2021	31-12-2025		€ -	€ 0	€ 40.000,00		€ 160.000,00		€ 160.000,00		€ 160.000						€ 520.000	€ 0	€ 520.000,00
TKI Life Sciences & Health - Heartchip II	Volledige projectkosten		1-10-2019	30-9-2023		€ -	€ 0	€ 30.749,00		€ 92.224,00										€ 122.973	€ 0	€ 122.973,00
Naam 4		-				€ -	€ 0													€ 0	€ 0	€ 0,00
						Totaal		€ 532.176,05		€ 1.596.300,62		€ 801.170,52		€ 160.000,00						€ 3.089.647,18	osten matching	

2.

Kosten van materialen en hulpmiddelen:				2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	Materiaal / hulpmiddel		Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	In cash	In kind	in cash	in kind	
Onderzoeksorganisaties																		
Organisatie 1,2 of 3 aanvraag	Organisatie congressen nationaal/internationaal		€ 25.000			€ 5.000		€ 5.000		€ 5.000		€ 5.000		€ 5.000		€ 25.000	€ 0	€ 25.000,00
Organisatie 1,2 of 3 aanvraag	Seed grants (4* 2.5k/jaar)		€ 50.000			€ 10.000		€ 10.000		€ 10.000		€ 10.000		€ 10.000		€ 50.000	€ 0	€ 50.000,00
																€ 0	€ 0	€ 0,00
Organisatie 4																€ 0	€ 0	€ 0,00
Ondernemingen																		
Onderneming 1																€ 0	€ 0	€ 0,00
Onderneming 2																€ 0	€ 0	€ 0,00
Onderneming 3																€ 0	€ 0	€ 0,00
Onderneming 4																€ 0	€ 0	€ 0,00
Overige (private) partners																		
Naam 1																€ 0	€ 0	€ 0,00
Naam 2																€ 0	€ 0	€ 0,00
Naam 3																€ 0	€ 0	€ 0,00
Naam 4																€ 0	€ 0	€ 0,00
		Totaal	€ 75.000													Totaal kosten materialen en hulpmiddelen		€ 75.000

3.

Kosten van gebruik van machines en apparatuur							2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	Machine / apparatuur			Prijs per gebruikseenheid	Gebruiks eenheden	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																					
Organisatie 1																			€ 0	€ 0	€ 0,00
Organisatie 2																			€ 0	€ 0	€ 0,00
Organisatie 3																			€ 0	€ 0	€ 0,00
Organisatie 4																			€ 0	€ 0	€ 0,00
Ondernemingen																					
Onderneming 1																			€ 0	€ 0	€ 0,00
Onderneming 2																			€ 0	€ 0	€ 0,00

Partner die de kosten maakt	Naam derde, omschrijving kosten	Kosten	2022 in cash	2022 in kind	2023 in cash	2023 in kind	2024 in cash	2024 in kind	2025 in cash	2025 in kind	2026 in cash	2026 in kind	2027 in cash	2027 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderneming 3															€ 0	€ 0	€ 0,00
Onderneming 4															€ 0	€ 0	€ 0,00
Overige (private) partners																	
Naam 1															€ 0	€ 0	€ 0,00
Naam 2															€ 0	€ 0	€ 0,00
Naam 3															€ 0	€ 0	€ 0,00
Naam 4															€ 0	€ 0	€ 0,00
	Totaal	**€ 0**														**Totaal kosten machines en apparatuur**	**€ 0**

4. **Aan derden verschuldigde kosten**

Partner die de kosten maakt	Naam derde, omschrijving kosten	Kosten	2022 in cash	2022 in kind	2023 in cash	2023 in kind	2024 in cash	2024 in kind	2025 in cash	2025 in kind	2026 in cash	2026 in kind	2027 in cash	2027 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderzoeksorganisaties																	
Organisatie 1															€ 0	€ 0	€ 0,00
Organisatie 2															€ 0	€ 0	€ 0,00
Organisatie 3															€ 0	€ 0	€ 0,00
Organisatie 4															€ 0	€ 0	€ 0,00
Ondernemingen																	
Onderneming 1															€ 0	€ 0	€ 0,00
Onderneming 2															€ 0	€ 0	€ 0,00
Onderneming 3															€ 0	€ 0	€ 0,00
Onderneming 4															€ 0	€ 0	€ 0,00
Overige (private) partners																	
Naam 1															€ 0	€ 0	€ 0,00
Naam 2															€ 0	€ 0	€ 0,00
Naam 3															€ 0	€ 0	€ 0,00
Naam 4															€ 0	€ 0	€ 0,00
	Totaal	**€ 0**														**Totaal kosten derden**	**€ 0**

5. **Publicatie, reis- en verblijfkosten**

Partner die de kosten maakt	Naam derde, omschrijving kosten	Kosten	2022 in cash	2022 in kind	2023 in cash	2023 in kind	2024 in cash	2024 in kind	2025 in cash	2025 in kind	2026 in cash	2026 in kind	2027 in cash	2027 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderzoeksorganisaties																	
Organisatie 1,2 of 3 aanvraag	benchfee, reis, en publicatiekosten CIFIC PD fellows	€ 56.300			€ 11.260		€ 11.260,00		€ 11.260		€ 11.260,00		€ 11.260,00		€ 56.300	€ 0	€ 56.300,00
Organisatie 2																€ 0	€ 0,00
Organisatie 3																€ 0	€ 0,00
Organisatie 4																€ 0	€ 0,00
Ondernemingen																	
Onderneming 1															€ 0	€ 0	€ 0,00
Onderneming 2															€ 0	€ 0	€ 0,00
Onderneming 3															€ 0	€ 0	€ 0,00
Onderneming 4															€ 0	€ 0	€ 0,00
Overige (private) partners																	
Naam 1															€ 0	€ 0	€ 0,00
Naam 2															€ 0	€ 0	€ 0,00
Naam 3															€ 0	€ 0	€ 0,00
Naam 4															€ 0	€ 0	€ 0,00
	Totaal	**€ 56.300**														**Totaal kosten derden**	**€ 56.300**

6. **Totale kosten** — **€ 1.999.954** — **€ 1.999.954**

FINANCIERING - Bijdragen per partner per jaar

	in Euro	2022 in cash	2022 in kind	2023 in cash	2023 in kind	2024 in cash	2024 in kind	2025 in cash	2025 in kind	2026 in cash	2026 in kind	2027 in cash	2027 in kind	Totaal in cash	Totaal in kind	Totaal per partner
Onderzoeksorganisaties																
1	Organisatie 1 TUD		€ 23.004		€ 92.015		€ 92.015		€ 92.015		€ 92.016		€ 69.011	€ 0	€ 391.065	€ 391.065
2	Organisatie 2 EMC													€ 0	€ 0	€ 0
3	Organisatie 3 Erasmus Universiteit													€ 0	€ 0	€ 0
…	Organisatie 1,2 of 3													€ 0	€ 0	€ 0
	Totale bijdrage onderzoeksorganis	€ 0	€ 23.004	€ 0	€ 92.015	€ 0	€ 92.015	€ 0	€ 92.015	€ 0	€ 92.016			€ 0	€ 391.065	€ 391.065
Ondernemingen																
4	Onderneming 1													€ 0	€ 0	€ 0
5	Onderneming 2													€ 0	€ 0	€ 0
6	Onderneming 3													€ 0	€ 0	€ 0
…	Onderneming 4													€ 0	€ 0	€ 0
	Totale bijdrage ondernemingen	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0			€ 0	€ 0	€ 0
Overige (private) partners																
7	Diverse matching uit bestaande pro	€ 532.176		€ 1.596.301		€ 801.171		€ 160.000		€ 0		€ 0		€ 3.089.647	€ 0	€ 3.089.647
8	Naam 2													€ 0	€ 0	€ 0
9	Naam 3													€ 0	€ 0	€ 0
…	Naam 4													€ 0	€ 0	€ 0
	Totale overige bijdrage	€ 532.176	€ 0	€ 1.596.301	€ 0	€ 801.171	€ 0	€ 160.000	€ 0	€ 0	€ 0			€ 3.089.647	€ 0	€ 3.089.647
		in cash	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**					**Totaal**
	Totaal budget per jaar	€ 532.176	€ 23.004	€ 1.596.301	€ 92.015	€ 801.171	€ 92.015	€ 160.000	€ 92.015	€ 0	€ 92.016					€ 3.480.712
																Funding gap
																1.480.758
	Totaal budget	€ 555.180		€ 1.688.316		€ 893.186		€ 252.015		€ 92.016				€ 3.480.712		

Delft, 8th of March 2022

DELMIC BV
Kanaalweg 4
2628EB, Delft

LETTER OF INTENT for the CIFIC Convergence Flagship PROJECT

Dear prof Herder,

I, Sander den Hoedt, in my capacity of CEO at Delmic B.V. hereby confirm that Delmic B.V. via its business units Fast Imaging B.V. and Delmic Cryo B.V. intends to contribute to the CIFIC Convergence Flagship project, as applied for by the main applicant, Jacob Hoogenboom, associate professor and section head at the Imaging Physics department of TU Delft.

Delmic develops innovative products for multi-modal, large-scale, and high-resolution (cryogenic) microscopy in life sciences and biomedicine. We have long-standing successful collaborations with the Imaging Physics department at TU Delft as well as an extensive network of microscopy end users in life sciences research institutes or university medical centres, which has helped us launch several new microscopy products in only a few years. We emphasise the importance of close collaborations between all these parties in microscopy innovation and development and we recognize the unique opportunity for a local convergent ecosystem combining facility, innovation, and product development to further accelerate the application and market introduction of revolutionary new microscopes for research and diagnosis.

Delmic currently contributes with in-cash and in-kind contributions to the NWO funded projects Cryo3beams (100k€ in-cash plus 144k€ in-kind; TUD applicants Hoogenboom & Jakobi) and Building Biomolecular Google Maps (60k€ in-cash; TUD applicants Carroll & Hoogenboom) which fully align with the CIFIC research agenda. We have further expressed our intent to contribute to the NWO Perspectief proposal on '3D nanoscale imaging, sensing, and control' (TUD main applicant Stallinga, EMC co-applicant Houtsmuller).

Delmic intends to support future CIFIC project proposals with similar in cash and in-kind contributions We expect CIFIC to accelerate project applications and microscopy developments particularly in the areas of super resolution correlative microscopy and large-scale microscopy and are more than willing to contribute to its success. In addition, we also foresee unique opportunities for novel clinical applications, such as the application of multi-beam FAST-EM at TU Delft for pathology for which we could contribute with our expertise.

Yours sincerely,

Yours sincerely,

Sander den Hoedt
CEO, DELMIC BV
8th of March 2022, Delft

DENSsolutions
Informaticalaan 12
2628ZD Delft
The Netherlands

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E. info@DENSsolutions.com

www.DENSsolutions.com

Date: 10 March 2022

LETTER OF INTENT
for the
CIFIC Convergence Flagship PROJECT

Dear prof Herder,

I, Dr. Hugo Pérez-Garza, in my capacity of CEO at DENSsolutions B.V. hereby confirm that DENSsolutions intends to contribute to the CIFIC Convergence Flagship project, as applied for by the main applicant, Jacob Hoogenboom, associate professor and section head at the Imaging Physics department of TU Delft.

DENSSolutions is a Delft-based company focusing on the development of innovative solutions for in-situ Transmission Electron Microscopy, using Micro Electromechanical Systems (MEMS) as our core competence. Together with the MEMS, we also develop and provide the sample holder, software, electronics and external hardware components to control the full experiment. Currently we have a comprehensive product portfolio and a global distribution network, making DENSsolutions a global leader in the in-situ TEM market. DENSSolutions is interested in the development and application of sample holders for high-resolution electron microscopy of biological macromolecules under native (liquid) or near-native (cryogenic) conditions.

We strongly support the collaboration between microscopy development and biomedical research oriented application within the CIFIC project. We have discussed several new project ideas with dr Arnab Ray Chaudhuri and hereby express our interest to support future projects with dr Chaudhuri and colleagues within CIFIC with cash or in-kind contributions if needed.

Yours sincerely,

Dr. Hugo Pérez-Garza

Chief Executive Officer (CEO)

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e hugo.perez@DENSsolutions.com

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we deliver

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Thermo Fisher
SCIENTIFIC
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Thermo Fisher Scientific
Materials & Structural Analysis
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5651 GH Eindhoven
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March 14, 2022

Subject : Support letter for the CIFIC Convergence Flagship project

Dear Prof. Herder,

With this letter we express our enthusiasm for the proposed project "Convergence Imaging Facility and Innovation Centre (CIFIC)" for which a full application is going to be submitted by Dr. Jacob Hoogenboom, Delft University of Technology. Thermo Fisher is interested to participate in the project and interact with the other project participants. Provided this project will be granted, Thermo Fisher will consider in which way it can contribute to the success of the project.

Thermo Fisher (Materials and Structural Analysis Division, MSD) is a world-leading supplier of equipment for analyzing, measuring and modifying sub-micron and atomic structures in the markets of materials research, nano-technology, life sciences and micro-electronics. Thermo Fisher designs, manufactures and supports a broad range of high-performance microscopy workflow solutions that provide images and answers at the micro-, nano- and picometer scales. Equipment includes transmission electron microscopes (TEM), scanning electron microscopes (SEM) and DualBeams™, which combine a SEM with a focused ion beam.

Thermo Fisher has a long history in collaboration with the Imaging Physics group at the TU Delft. One of the most recent collaborative projects concerns the development of a MultiBeam SEM which will enable much higher throughput of nm resolution imaging of biological samples. Thermo Fisher recognizes the added value of close interaction with microscopy facility staff to target workflow challenges in the implementation of novel microscopy techniques in life sciences and health.

Best regards,

Dr. A.F. de Jong
Director, Strategic programs
Thermo Fisher Scientific, MSD
Eindhoven, Netherlands

Cc: Dr. Jacob Hoogeboom,section head, Imaging Physics Department, TU Delft

SINGLE QUANTUM
Excellence in photon detection

LETTER OF INTENT
for the CIFIC Convergence Flagship PROJECT

Date: 11 March 2022

Dear Prof. Herder,

I, S.N. Dorenbos, in my capacity of CEO at Single Quantum BV hereby confirm that Single Quantum BV intends to contribute to the CIFIC Convergence Flagship project, as applied for by the main applicant, Jacob Hoogenboom, associate professor and section head at the Imaging Physics department of TU Delft.

Single Quantum is a 2012 TU Delft spin-off developing highly sensitive single photon detectors based on superconducting nanowires. Our company has grown considerably over the past ten years and our products have been implemented by more than 100 industrial and academic laboratories all over the world to enable complex optical measurements. Single Quantum is highly interested in extending the use case of its detectors into the domain of life sciences and biomedical microscopy. We have a long standing collaboration with dr Iman Zadeh, tenure-track PI at the Imaging Physics department of TU Delft. Via dr. Zadeh, we are now discussing a collaboration on multi-photon detection for brain imaging with dr. Brinks and we are interested in investigating further applications in, e.g., light-sheet, super-resolution, and correlative light and electron microscopy with the CIFIC consortium. We therefore support the CIFIC proposal and would be willing to support future CIFIC-related research proposals with in-kind or potentially in-cash contributions.

Yours sincerely,

S.N. Dorenbos

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Technolution

Delft University of Technology
Attn Dr. ir. Jacob Hoogenboom
Department of Imaging Physics
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2628 CJ Delft
The Netherlands

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E info@technolution.com
I www.technolution.com

Company no. 29031050
ISO 9001 certified

Subject: Letter of Intent for CIFIC Convergence Flagship Project

Place:	Gouda
Date:	11 March 2022
Reference:	SVO/GRA.22011

Dear Mr. Hoogenboom,

I, Serge de Vos, in my capacity of CTO at Technolution B.V. hereby confirm that Technolution B.V. intends to contribute to the CIFIC Convergence Flagship project, as applied for by the main applicant, Jacob Hoogenboom, associate professor and section head at the Imaging Physics department of TU Delft.

Technolution has successfully developed an innovative multi-beam electron microscope (FAST-EM) in a consortium with TU Delft (Kruit and Hoogenboom), Delmic and Thermo Fisher Scientific. Technolution currently develops an imaging and data platform for large-scale microscopy data together with TU Delft researchers Hoogenboom and Carroll in the framework of the EFRO-React funded IMDAP project (2 PD positions for 2 years). We are also coordinating the metrology roadmap in the NXTGEN HIGHTECH proposal for the Nationaal Groeifonds in which 1M€ funding is included for fundamental science projects in multi-beam electron microscopy and super-resolution light microscopy technology at TU Delft (with Smith and Hoogenboom). Collaborations within CIFIC would greatly benefit these projects and allow us to extent use cases for our products to the biomedical and clinical domain.

Technolution intends to support future CIFIC projects via joint applications like the examples mentioned above, by involving CIFIC researchers in industrial-oriented funding schemes coordinated by the company, and/or by providing in-cash and in-kind contributions to NWO or EU co-funded project proposals.

Yours sincerely,
Technolution B.V.

Serge de Vos
CTO

Redefining
solutions

Date: 11 March 2022
Betreft: Convergence Flagship CIFIC

Prof.dr. S. Stallinga
Department of Imaging Physics
Faculty of Applied Sciences

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2628 CJ Delft

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E: s.stallinga@tudelft.nl

To whom it may concern,

Herewith I want to give my full, wholehearted support for the health-tech convergence proposal for a Flagship on Convergence Imaging Facility and Innovation Centre (CIFIC), led by dr.ir. J.P. Hoogenboom, associate professor and section head in my department of Imaging Physics at TU Delft. Bringing together technologists and end-users in the clinic of a range of microscopy workflows will boost research into disease models and will advance diagnosis and treatments for a variety of diseases. A better match with the goals of health-tech convergence can hardly be envisioned.

My department intends to support CIFIC in kind by hiring dr. Miriam Menzel as new tenure-track PI, provided the job application process can be successfully concluded. Dr. Menzel will work in the area of label-free, scattering-based microscopy for clinical and live imaging applications, in line with the goals of this Flagship proposal.

My hope is that the initiative of this consortium will receive the support from our universities to accelerate and sustain its growth to international leadership in co-development of microscopy technology and clinical application.

With kind regards,

prof. dr. Sjoerd Stallinga
Head of Department of Imaging Physics, faculty of Applied Sciences, TU Delft

Annex IV – Selected references of CIFIC PIs

TU DELFT

Smith

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ERASMUS MC

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Chien

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